

1. Backsavers
 - a. C = number of Collegiate backpacks produced per week
 - b. M = number of Mini backpacks produced per week
 - c. Z = Maximize total
2. Each Collegiate earns a profit of \$32 and each Mini earns a profit of \$24. Since management wants to maximize profit:
 - a. Maximize $Z=32C+24M$
3. The company receives 5,000 square feet of nylon each week. Each Collegiate uses 3 square feet and each Mini uses 2 square feet.
 - a. $3C+2M \leq 5000$
 - b. The sales forecast says that no more than 1,000 Collegiates can be sold each week: $C \leq 1000$
 - c. No more than 1,200 Minis can be sold each week: $M \leq 1200$
 - d. Each Collegiate requires 45 minutes of labor and each Mini requires 40 minutes. There are 35 workers who each provide 40 hours per week.
 - e. Total labor available: $35(40)(60)=84,000$ minutes
 - f. So the labor constraint is: $45C+40M \leq 84,000$
 - g. Finally, production cannot be negative: $C, M \geq 0$
4. Maximize $Z=32C+24M$
 - a. Subject to:
 - b. $3C+2M \leq 5000$
 - c. $45C+40M \leq 84,000$
 - d. $C \leq 1000$ $M \leq 1200$
 - e. $C, M \geq 0$

1.
 - a. L_1, L_2, L_3 = number of large units produced at Plants 1, 2, and 3
 - b. M_1, M_2, M_3 = number of medium units produced at Plants 1, 2, and 3
 - c. S_1, S_2, S_3 = number of small units produced at Plants 1, 2, and 3
 - d. u = percentage of excess production capacity used at each plant
2. The profit per unit is: Large = \$420, Medium = \$360, Small = \$300
 - a. Maximize $Z=420(L_1+L_2+L_3)+360(M_1+M_2+M_3)+300(S_1+S_2+S_3)$
3. Plants 1, 2, and 3 can produce at most 750, 900, and 450 total units per day.
 - a. $L_1+M_1+S_1=750u$
 - b. $L_2+M_2+S_2=900u$
 - c. $L_3+M_3+S_3=450u$
4. Each large unit requires 20 square feet, each medium unit requires 15 square feet, and each small unit requires 12 square feet. The available storage is 13,000 square feet at Plant 1, 12,000 at Plant 2, and 5,000 at Plant 3.

- a. Plant 1: $20L_1+15M_1+12S_1\leq 13,000$
 - b. Plant 2: $20L_2+15M_2+12S_2\leq 12,000$
 - c. $20L_3+15M_3+12S_3\leq 5,000$
- 5. The company can sell at most 900 large units, 1,200 medium units, and 750 small units per day.
 - a. Large: $L_1+L_2+L_3\leq 900$
 - b. Medium: $M_1+M_2+M_3\leq 1200$
 - c. Small: $S_1+S_2+S_3\leq 750$
- 6. Full Formulation: Maximize $Z=420(L_1+L_2+L_3)+360(M_1+M_2+M_3)+300(S_1+S_2+S_3)$

$$L_1+M_1+S_1=750u$$

$$L_2+M_2+S_2=900u$$

$$L_3+M_3+S_3=450u$$

$$20L_1+15M_1+12S_1\leq 13,000$$

$$20L_2+15M_2+12S_2\leq 12,000$$

$$20L_3+15M_3+12S_3\leq 5,000$$

$$L_1+L_2+L_3\leq 900$$

$$M_1+M_2+M_3\leq 1200$$

$$S_1+S_2+S_3\leq 750$$

$$L, M, S \geq 0$$

$$0 \leq u \leq 1$$